Izabella （Zhiqi） Yu

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Motivated and detail-oriented 3rd-year Computer Engineering student with good multi-tasking and collaboration skills, 3+ years of experience designing and analyzing computer components and systems. Proficient in C/C++, Python, and SQL. Seeking an opportunity of 12-16 months (ends before Sep.2026) to apply technical skills and problem-solving abilities, with a passion for working in a dynamic and challenging environment.

Education

Bachelor of Applied Science and Engineering (B.A.Sc) in Computer Engineering + PEY Co-op University of Toronto
Toronto, Ontario Sep.2021 - Apr.2027
Minor in Artificial Intelligence

Relevant Courses:

Introduction to Machine Learning	Applied Fundamentals of Deep Learning	Software Engineering
Algorithms and Data Structures	Introduction to Database	Calculus I&II&III
Software Communication & Design	Probability and Applications	Computer Organization

Skills:

Coding languages and relative Proficiency:

C/C++	Proficient	Python	Proficient	SQL	Proficient
HTML/CSS	Proficient	JavaScript	Intermediate	Java	Rudimentary

Programming/ Frameworks	PyTorch, TensorFlow, NumPy, Scikit-learn, PostgreSQL, SQLite, Django, MATLAB
Development Environments	VScode, Arduino IDE, Clion, PyCharm, Quartus, Google Colab
Version Control/Management	Git/GitHub, JIRA, Confluence
Professional Skills	Object-Oriented Design, Research, Risk Management, Teamwork, Complexity Analysis
Hardware Design	Verilog, Nios II, FPGA, Arduino
CAD /Simulation	Fusion360, LTSpice, ModelSim

Project:

Identify Malignant Melanoma Skin Lesions Through A CNN (AI - Deep Learning Model) Jan.2025 - Present
University of Toronto, Toronto, ON
Project from the University of Toronto course APS360 (Applied Fundamental of Deep Learning)

- Collaborated in a team of 4 to design, build, and train a Convolutional Neural Network for a specific task.
- Built and trained the model using Google Colab Notebook, PyTorch, TensorFlow, and Scikit-learn.
- Improved the model and tuning hyper-parameters to try getting higher accuracy than in existing reports.

“UofTrade” (A web-based Second-Hand Hub) Sep. - Dec.2024
University of Toronto, Toronto, ON
Project from the University of Toronto course ECE444 (Software Engineering)

- Collaborated in a team of 6 as a Backend Developer to implement a second-hand exchange platform.
- Started from scratch, applied Agile development methodology, designed the User Interface & optimized algorithms.
- Built a web application’s backend using Django framework, applied Python and PostgreSQL.

"Pipe Connection" (Hardware Game on FPGA board) Mar. - Apr.2024
University of Toronto, Toronto, ON
Project from the University of Toronto course ECE243 (Computer Organization)

- Designed and implemented an interactive FPGA-based puzzle game with increasing complexity and audiovisual features.
- Developed robust pipe-connectivity algorithms, adaptable for future applications & different hardware platforms.

- Enhanced teamwork, FPGA prototyping, problem-solving, and C programming skills through close collaboration and meticulous attention to detail in a group of 2.

A transit-centric map Jan. - Apr.2024

University of Toronto, Toronto, ON

Project from the University of Toronto course ECE297 (Software Communication & Design)

- Collaborated in a team of 3 as a Full-stack Developer to implement a GIS-based map in C++.
- Started from scratch, designed the User Interface, and implemented and optimized algorithms.
- Built a usable map with basic path-finding and multi-destination route-planning functionality.

Line-tracking robot Jun.2023

University of Toronto, Toronto, ON

8-day Project from the University of Toronto 2023 Electrical & Computer Engineering robotic workshop.

- Learned how to use Fusion360, Arduino IDE, and basic knowledge of I/O devices.
- Cooperated with one teammate to design the robot and adjust the sensors' layout.
- Optimize the Arduino codes' algorithm to increase stability, accuracy, and speed of line-tracking
- Achieved the highest accuracy and stability and third-quickest place in the workshop's final competition.

Experiences:

Vice-President & Co-Founder of Physics and Astronomy Club Sep.2018 - Jun.2020

Guanghua Cambridge International School, Shanghai, China

- Bring students who are interested in Physics and Astronomy together and build a communication platform.
- Organized programs and research about Physics and Astronomy for students.
- Encourage and support students to attend competitions and challenges related to the Physics or Astronomy area.